Data Structures

BS (CS) _Fall_2024

Lab_09 Tasks

Learning Objectives:

1. Stack

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB09* **e.g. i23XXXX_A_LAB07(Also name your cpp file like this.)**
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don’t follow the above-mentioned submission instruction, you will be marked **zero.**
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Question 1

You are given a maze represented as an N*M binary matrix of blocks, and a rat is initially located at (0, 0) in the maze, denoted as maze[0][0]. The objective of the rat is to reach a specific location in the maze, represented by the coordinates (fx, fy), where there is food.
In the maze matrix, each cell is either a 0 (indicating a dead end) or a 1 (indicating a valid path). The rat can move in any of the four cardinal directions (up, down, left, or right) but cannot move diagonally.
Your task is to write a C++ program that checks if there exists a path for the rat to reach the food or not. You do not need to print the path itself, just determine if it's possible for the rat to reach the food.

You will need to make the following c:

CELL (class/struct):

The Cell class is defined to represent a cell in the maze. It contains the x and y coordinates of the cell and the dir variable to store the direction the search is currently moving in.

VISITED (2d array)

An array to keep track of visited cells.)

IsReachable (a function)

This function isReachable takes the maze as input and returns a boolean indicating whether a. path from the start point to the target point exists. In this function you need to do the following things.

1. Initializes a stack and pushes the starting cell onto the stack.
2. The visited array is initialized with all cells marked as visited (set to true).
3. Inside the main loop (while stack not empty), it checks if the current cell is the target cell (fx and fy). If it is, the function returns true, indicating that a path is found. If not, it examines the neighboring cells in different directions (up, left, down, right) and pushes unvisited valid cells onto the stack, updating the visited array to mark them as visited

Question 2

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:

- Open brackets must be closed by the same type of brackets.
- Open brackets must be closed in the correct order.
- Every close bracket has a corresponding open bracket of the same type